Helical attractors on contact 3-manifolds: basin geometry, Whitney singularities, and rotating electromagnetic applications

Pablo Nogueira Grossi
G6 LLC, Newark, New Jersey, USA · grossiatwork@gmail.com · ORCID: 0009-0000-6496-2186
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We study a three-dimensional ODE on the contact 3-manifold $(\mathbb{R}^3,\, \alpha = dz - r^2\, d\theta)$ in cylindrical coordinates, where $r^2\, d\theta$ encodes angular momentum as a one-form. The system is $\dot{r} = r(1-r^2) + \varepsilon(r-1)e^{-z}$, $\dot{\theta} = 1, \dot{z} = r^2 - \varepsilon(r-1)^2 e^{-z}$, $\varepsilon = 2$. The contact condition $\alpha = 0$ reads $dz = r^2\, d\theta$, identifying vertical lift directly with angular momentum — a geometric identity rather than a force law. We prove that the unit helix $\Gamma = \{r = 1, \dot{\theta} = 1, \dot{z} = 1\}$ is a globally attracting limit cycle for all $r(0) > r^*$ and $z(0) \geq \log 2$, exponential convergence rate $\mu = -2$.

Principal certified result. The inner basin boundary r^* is located via adaptive bisection with the DOP853 integrator (rtol $= 10^{-12}$, atol $= 10^{-14}$): $r^* = 0.77594059$ (certified to 7 significant figures; reproducible on any IEEE-754 platform, `certify_rstar.py` [2]). This lies strictly above the Lyapunov bound $2/3 \approx 0.667$. Full stability hierarchy: $\varepsilon_0 = \frac{1}{3} < \frac{2}{3} < r^* = 0.77594 < \kappa^* = \sqrt{7/9} \approx 0.882 < 1$, where $\varepsilon_0 = \frac{1}{3}$ is the Lyapunov stability radius (separate from $\det J$).

Whitney A_1 singularity. We identify r^* as a Whitney A_1 (standard fold) singularity of the asymptotic radial map $\Phi : \mathbb{R}_+ \to \{1\}$. Trajectories from the outer basin ($r > 1$) and inner basin ($r^* < r < 1$) follow qualitatively distinct convergence paths that fold onto the single attractor $r = 1$. In a rotating electromagnetic realisation the fold boundary corresponds to a corona discharge threshold — a structural consequence of the contact geometry, not a fit parameter.

Closed-form saddle result (Theorem B.3). The Jacobian at the saddle (r^*, z^*) satisfies, proved in closed form in [3]:

$$\operatorname{tr}(J)\big|_{\text{saddle}} = 2\cos\left(\tfrac{2\pi}{7}\right) \approx 1.247, \qquad \det(J) = r_s^2(-1 - r_s - 2r_s^2) \approx -0.364,$$

The value $2\cos(2\pi/7)$ is the largest real root of $x^3 + x^2 - 2x - 1 = 0$, the minimal polynomial of $\operatorname{Re}(e^{2\pi i/7})$. Here $r_s = 2\cos(3\pi/7) \approx 0.4450$ is the saddle coordinate (root of $x^3 - x^2 - 2x + 1 = 0$); the negative determinant certifies the saddle. The Lyapunov stability radius $\varepsilon_0 = \frac{1}{3}$ is a *separate* quantity.

Framework and open obligation. The system sits at the Δ (tetranacci) rung of the n-bonacci recurrence ladder (dominant root $\approx 1.927 \to \tau = 2$) within the dm^3 / Principia Orthogona operator chain $G = U \circ F \circ K \circ C$, $\varepsilon = \tau = 2$. Global attractivity, the Whitney identification, and $\operatorname{tr}(J) = 2\cos(2\pi/7)$ are all established. The single remaining open problem is the analytic characterisation of r^* in closed form. Lean 4 mechanisation is in progress (AXLE, `github.com/TOTOGT/AXLE`); all results, proofs, and reproducibility scripts are available for independent verification at `totogt.github.io/3M/law3m.html`.

Keywords: contact geometry · helical attractor · Whitney A_1 singularity · basin of attraction · rotating electromagnetic system · Lyapunov stability

[1] P.N. Grossi, *Principia Orthogona*, doi:10.5281/zenodo.19117399 (2024–2026).
[2] P.N. Grossi, `certify_rstar.py`, github.com/TOTOGT/3M (2026).
[3] P.N. Grossi, *Contact-Geometric Theory of Generative Transitions: Seven Proofs of the Tribonacci Constant*, doi:10.5281/zenodo.20682934 (2026).